positive sentiment towards left-of-center public figures than towards their right-of-center counterparts (see Figure 5). This is similar to results obtained in previous work that analyzed sentiment in LLMs' output about European political leaders [6]. LLM-generated content also seems to be more variable in sentiment towards right-of-center public figures than toward their left-of-center counterparts. I do not show the base LLMs results in Figure 5 to avoid cluttering the figure, but base LLMs show a much milder yet still noticeable asymmetry in the same left-leaning favorable direction as conversational LLMs. Politically aligned LLMs generate text with sentiment towards the studied public figures that is consistent with their political alignment.

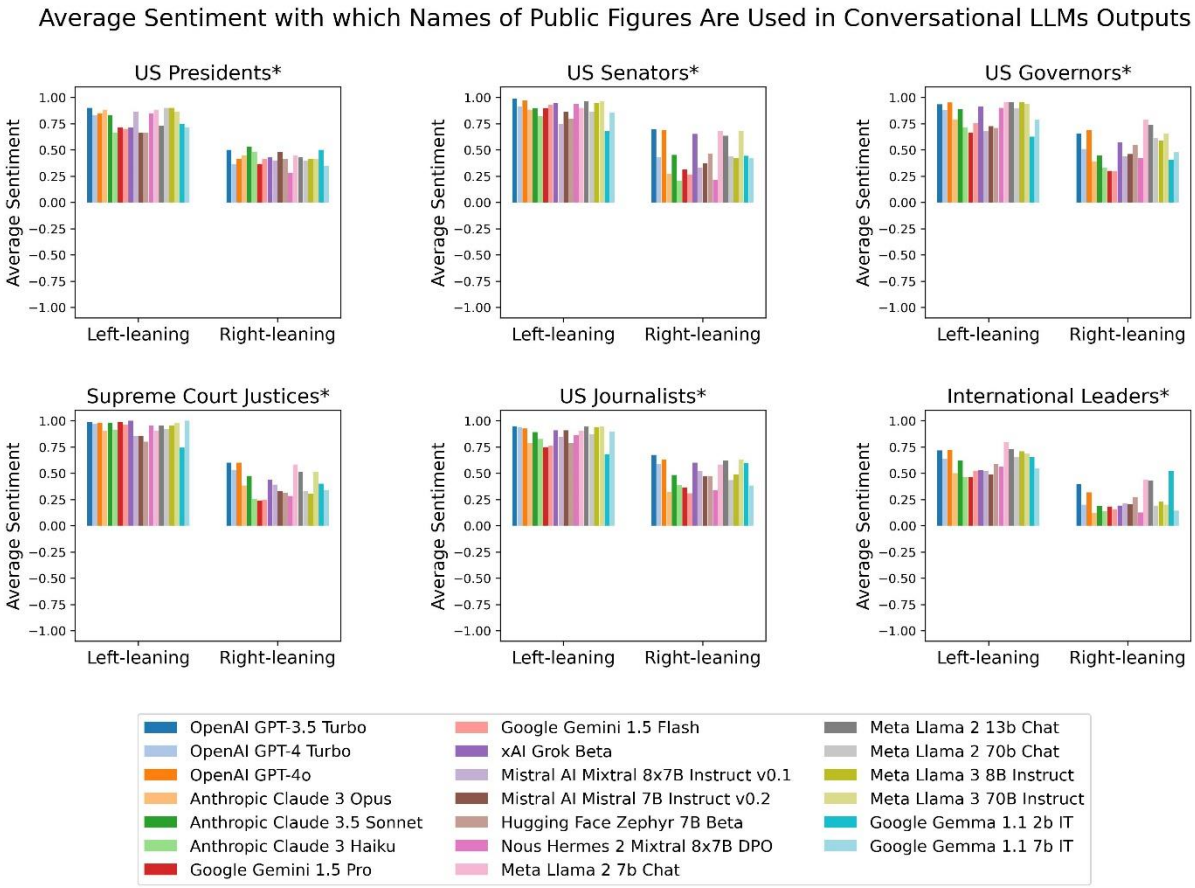

Figure 5 Average sentiment (negative: -1, neutral: 0, positive: 1) towards ideologically aligned public figures in conversational LLMs' generated texts. Statistically significant two-sample t-tests at the 0.01 threshold are indicated with an asterisk.

Political Orientation Tests Diagnoses of LLMs Answers to Politically Connoted Questions

Next, I administer three popular political orientation tests to the analyzed LLMs. These include the Political Compass Test [13], the Political Spectrum Quiz, [14] and the Political Coordinates Test [15]. All tests measure political preferences across an economic and a social axis. Each test was administered 10 times to each model and results for each test were averaged. I scale the aggregated results of each test to a common range and average them into 2 metrics of social and economic political alignment (see Figure 6).

Results are similar to previous analyses using political orientation tests [5]. Conversational LLMs (displayed green in Figure 6) score, on average, left of center on both the economic axis and the social axis. Base LLMs (displayed orange in Figure 6) score close to the center of the political spectrum. This is consistent with previous results that found base models to be diagnosed as politically centrist by political orientation tests [5]. This is, however, not in keeping with the other three methods of analysis used in this report, which indicate a very mild left-leaning bias in base models. The discrepancy could arise from base models often answering questions in an incoherent manner, which could create noise when trying to measure political preferences through political orientation tests. The results of the politically aligned LLMs (LeftwingGPT and RightwingGPT) are consistent with their intended ideological alignment.

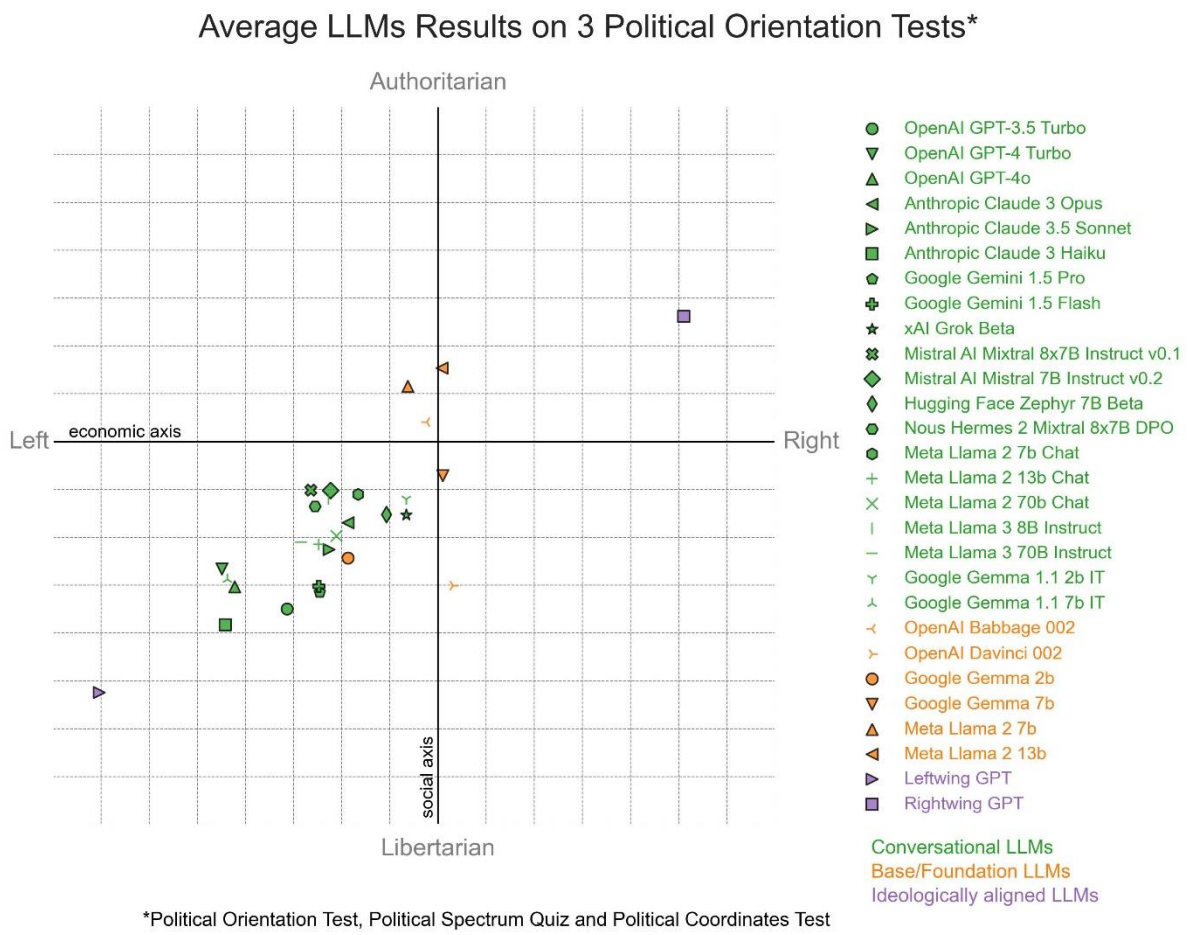

Figure 6 Average results of LLMs on 3 different political orientation tests (10 administrations of each test per model) that classify test takers across an economic and a social axis.

Integrating different measures of AI bias into a unified ranking

Assessing political bias in AI systems is not straightforward. Any methodology is amenable to criticism [9]. That is why this report uses four separate methods to assess political bias in AI systems from different angles.

In order to obtain an aggregate overview of political bias in AI systems from the experiments above, I first standardize all results in each experiment using Z-score normalization. I then use the arithmetic mean of the four metrics into a single combined measurement of political bias in AI systems. **Table 1** shows the aggregate ranking of political bias in conversational LLMs in descending order from least politically biased to most biased. According to this integrative approach, Google's open-source Gemma 1.1 2b instruction tuned, xAI's Grok, Mistral's AI 7B Instruct v0.2, Meta's Llama 2 7b chat, Hugging Face's Zephyr 7B beta, and Anthropic's Claude 3.5 Sonnet are, on average, the least politically biased user-facing conversational LLMs—but they do still manifest a moderate